Assignment 4

Network Threats & Mitigation & Physical Security & Risk

1. **Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).**

No.	Network Threat	Real-World Example
1.	**Phishing**	An Instagram user receives a fake login link designed to steal their password.
2.	**DDoS Attack**	Attackers flood a shopping website like Flipkart with excessive requests, making it unavailable.
3.	**Man-in-the-Middle (MITM)**	An attacker on an insecure public Wi-Fi network attempts to intercept a user's unencrypted communication.
4.	**Malware**	A WhatsApp user downloads a malicious file that infects their device.
5.	**Password Attack**	An attacker repeatedly guesses a Paytm user's password to try to access their account.

2. **Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value. Constraint: The script should print the number of attempts it took to guess the correct PIN.**

➢ **Python Program**

```
# Preset 4-digit PIN
correct_pin = "2584"
attempts = 0

# Try all possible 4-digit combinations
for number in range(10000):
    pin = f"{number:04d}"
    attempts += 1

    if pin == correct_pin:
        print("PIN guessed:", pin)
        print("Number of attempts:", attempts)
        break
```

➢ **Output**

PIN guessed: 2584

Number of attempts: 2585

➢ **Explanation**

- The program checks every PIN from 0000 to 9999.
- It increases the attempt counter after each guess.
- When the correct PIN is found, it prints the PIN and total attempts, then stops.

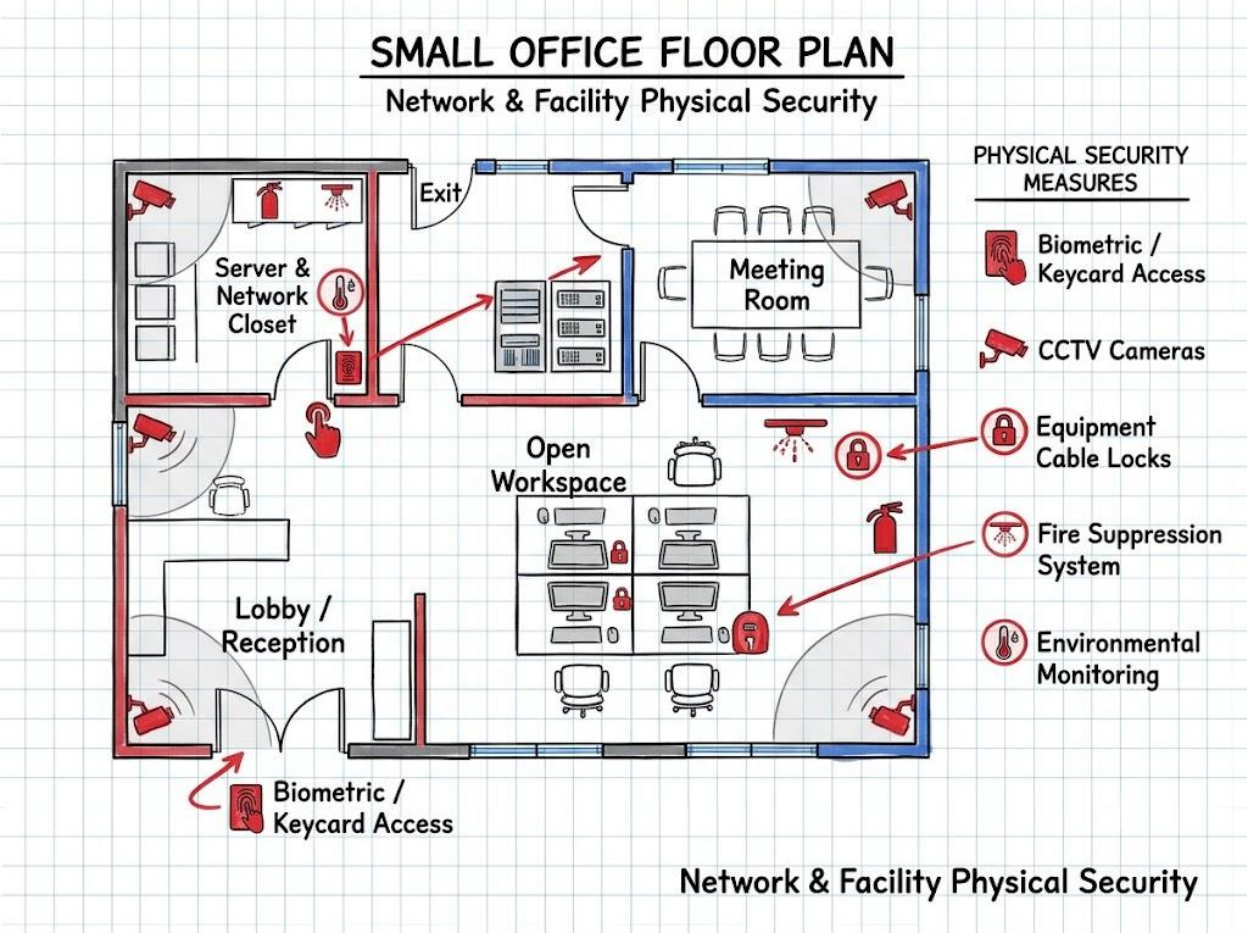
➢ **Result**

The program successfully simulates a brute-force attack and calculates how many attempts were needed to guess the correct 4-digit PIN.

3. **List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.**

No.	Physical Security Risk	Mitigation Step
1.	**Unauthorized Access:** An unauthorized person enters the server room.	Use access cards, biometric locks, and CCTV cameras.
2.	**Fire:** A fire damages servers and network equipment.	Install smoke detectors and fire-suppression systems.
3.	**Power Failure:** A power outage shuts down servers.	Use UPS systems and backup generators.
4.	**Overheating:** High temperatures damage server hardware.	Install proper cooling systems and temperature monitoring.

4. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.



1) **Biometric / Keycard Access:** Installed at the main entrance and the Server & Network Closet to restrict unauthorized entry.

2) **CCTV Cameras:** Strategically placed in corners and critical entrance points—especially facing the Server Closet and main lobby—for continuous visual monitoring.

3) **Equipment Cable Locks**: Securing valuable workstations and server hardware to prevent physical theft.

4) **Additional Measures:** Includes environmental monitoring for Server Room heat control and fire suppression systems (sprinklers/extinguishers) to prevent physical disasters.

5. **Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.**

1. ☐ **Use Firewalls:** Configure firewalls to block unauthorized network traffic and suspicious connections.

2. ☐ **Install Antivirus**: Keep antivirus software updated on all lab computers to protect against malware.

3. ☐ **Secure Passwords:** Use strong passwords and separate student and administrator accounts.

4. ☐ **Control Physical Access:** Lock the computer lab when not in use and allow access only to authorized students and staff.

5. ☐ **Protect Network Devices:** Keep routers, switches, and servers in a locked cabinet to prevent tampering.

6. ☐ **Back Up Data:** Regularly back up important college files and student project work to a secure location.

7. ☐ **Monitor and Update Systems:** Install security updates regularly and check system logs for suspicious activities**.**

Review and Edit: I adapted the checklist for a college computer lab by focusing on student accounts, lab access, network equipment, and student project data.

Result: These seven actions help reduce network threats and physical security risks in a college computer lab.